Two Additional Two-Factor Simulation Studies: MSE Results

This note reports the Monte-Carlo mean-squared-error (MSE) results for **two new simulation studies** that extend the single-factor study of the main paper to a **two-factor** functional structural equation model (FSEM). All quantities are computed from the stored simulation output (folders `outputs/` and `outputs3/`); **no simulation was re-run**. The tables are intended to be incorporated into the manuscript later.

Throughout, the latent factors are zero-mean Gaussian processes observed only through their indicators. Each replicate is summarised by the integrated squared error between the estimated and the true (standardised) functional parameter, and the MSE is the Monte-Carlo average of that error over the replicates in each design cell. We use the manuscript notation:

Symbol	Meaning
λ	factor-loading function (indicator on factor)
β	factor intercept function
γ	structural (latent-on-latent) regression function
$\phi,\ \nu$	leading eigenfunction / eigenvalue of a factor-error process
$\psi,\ \mu$	leading eigenfunction / eigenvalue of a latent-process
σ^2	measurement-error variance

Both studies use $J = 6$ indicators $(z_1, \ldots, z_6)$, a B-spline basis as in the main study, signal-to-noise ratio SNR $= 4$ and correlation parameter $\rho = 0.3$, with Matérn covariance for the latent processes. Loadings combine a *fixed-concurrent* anchor (z_1, z_4), *concurrent* effects (z_2, z_5, z_6) and a *historical* effect (z_3). Sample sizes $N \in \{50, 100\}$ and number of observation points $M \in \{10, 20\}$ are crossed. For readability all loading and intercept MSEs are reported $\times 10^{-2}$.

Study A — Two factors, *uncorrelated*

Design. Two latent factors η_1, η_2 are generated **independently** (no structural path between them). Indicators z_1, z_2, z_3 load on η_1 and z_4, z_5, z_6 load on η_2. Each factor has an intercept-only structural equation ($\eta_1 \sim 1$, $\eta_2 \sim 1$). This study checks that the estimator recovers a multi-factor measurement model and the (independent) latent-process covariances when there is no latent dependence to exploit. Results in `outputs/` comprise the four cells $N \in \{50, 100\} \times M \in \{10, 20\}$ with 80 replicates each.

Findings. Every loading and intercept MSE decreases monotonically as the sample size N grows and as the curves are observed more densely (M), confirming consistent estimation of the two-factor measurement model. The *historical* loading λ_3 is the hardest to estimate (largest MSE), as expected for a two-dimensional surface, but it also improves with N and M. The leading eigenfunctions/eigenvalues (ϕ_1, ν_1) of the factor-error processes and the latent-process covariances (ψ_1, μ_1) are recovered with small error (e.g. ψ_1, μ_1 MSE $\approx$ 1–5 $\times 10^{-3}$, all decreasing in N, M);

Table 1: Study A: factor-loading MSE λ ($\times 10^{-2}$).

N	M	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
50	10	2.46	3.15	10.18	6.32	6.41	5.22
50	20	1.82	2.14	8.72	3.79	3.77	3.71
100	10	1.36	2.09	8.88	3.75	3.97	3.06
100	20	1.25	1.52	7.62	1.85	1.98	1.90

Table 2: Study A: factor-intercept MSE β ($\times 10^{-2}$).

N	M	β_1	β_2	β_3	β_4	β_5	β_6
50	10	5.54	4.72	2.25	3.99	3.66	3.41
50	20	5.51	5.07	2.49	3.48	3.12	3.03
100	10	3.35	3.32	1.92	1.85	1.93	1.83
100	20	2.75	2.43	1.45	1.78	1.66	1.59

higher-order eigen-components are subject to the usual sign/ordering non-identifiability and are not tabulated. Because the factors are independent, no structural (γ) parameters are present.

Study B — Two factors, *dependent* (well- vs mis-specified)

Design. The two factors are now **dependent**: the data-generating model includes a structural path $\eta_1 \leftarrow \eta_2$ (a concurrent latent-on-latent regression γ), in addition to the same six-indicator measurement model (z_1, z_2, z_3 on η_1; z_4, z_5, z_6 on η_2). Two estimators are fitted to every data set:

- **Well-specified (`.w`)** — the fitted model includes the structural path $\eta_1 \sim \eta_2$ (the correct dependence).
- **Mis-specified (`.m`)** — the fitted model **omits** the path ($\eta_1 \sim 1$ only), so the latent dependence is ignored.

This study quantifies (i) how well the structural function γ is recovered and (ii) the cost of ignoring latent dependence. Results in `outputs3/` cover $N = 100$ with $M \in \{10, 20\}$.

Table 3: Study B: factor-loading MSE λ ($\times 10^{-2}$), well-specified vs mis-specified.

Fit	N	M	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
well	100	10	8.11	8.41	6.78	2.67	2.84	2.17
well	100	20	3.23	3.42	4.40	1.69	1.64	1.67
miss	100	10	68.63	73.12	75.55	3.81	4.31	3.00
miss	100	20	68.31	73.44	78.48	2.00	2.18	2.15

(The mis-specified fit contains no γ because the path is omitted.)

Findings. Under the **well-specified** model the loadings of all six indicators are estimated accurately and improve sharply with denser sampling ($M : 10 \to 20$ roughly halves the loading MSE). The structural function γ is the most challenging parameter, but its MSE falls by a factor

Table 4: Study B: structural-function MSE γ ($\eta_1 \leftarrow \eta_2$), well-specified.

N	M	γ
100	10	0.408
100	20	0.117

of ~ 3.5 (from 0.41 to 0.12) as M increases from 10 to 20, showing that the latent-on-latent effect is recovered well once the curves are reasonably sampled.

The **mis-specified** fit reveals the cost of ignoring the dependence: the loadings of the indicators of the *contaminated* factor η_1 (z_1, z_2, z_3) deteriorate by **more than an order of magnitude** ($\lambda_{1\text{–}3}$ MSE jumps from ≈ 0.03–0.08 to ≈ 0.69–0.78 and, tellingly, **does not improve with** M), whereas the loadings of η_2's indicators (z_4, z_5, z_6) are essentially unchanged. Omitting the structural path therefore biases exactly the part of the measurement model attached to the response factor, while leaving the covariate factor intact — a clean and interpretable failure mode.

Discussion

Across the original single-factor study and the two new two-factor studies the estimator behaves consistently:

- **Consistency.** In every correctly-specified setting — the one-factor model of the main paper, Study A (uncorrelated factors) and the well-specified Study B (dependent factors) — all functional MSEs decrease monotonically with both the sample size N and the sampling density M. The main study established this for a single factor; the new studies show it continues to hold when the measurement model carries two factors and when the factors are linked by a structural path.
- **Scalability to richer measurement models.** Study A demonstrates that the EM estimator handles several indicators per factor and multiple independent factors without loss of accuracy: loadings, intercepts and the leading latent-covariance eigen-components are all recovered with small error.
- **Latent dependence is recoverable but data-hungry.** In the well-specified Study B the structural function γ has the largest MSE of any parameter yet improves rapidly with M, mirroring the difficulty of the *historical* surface effect seen in the main study. Adequate sampling density is the key driver of accurate structural estimation.
- **Mis-specification is localised but severe.** Ignoring a true latent-on-latent path inflates the loading MSE of the affected factor by more than a factor of ten and removes the usual benefit of denser sampling, while leaving the unaffected factor's parameters intact. This underlines the practical importance of correctly specifying the structural part of the model and shows that the damage is confined to the response factor rather than propagating through the whole system.

Taken together, the two additional studies corroborate the conclusions of the main simulation — consistent, sampling-density-driven estimation — and extend them to multi-factor measurement models and to latent structural dependence, while quantifying the cost of structural mis-specification.

Coverage rates are not reported here; this note covers the MSE results only.